MATHS MANTHAN
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CBSE CLASS 10 | MATHEMATICS

SOME APPLICATIONS OF TRIGONOMETRY

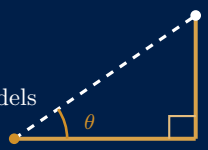
Heights & Distances Master Concept, Formula & Vector Geometry Handbook

Academic Year 2026–27

Visualize the Geometry. Anchor the Right Triangle.
Calculate with Precision.

Line of Sight | Elevation & Depression | 30°–45°–60° Engines | Canonical 2-Triangle Models

A publication-quality Maths Manthan learning and revision resource



What this handbook includes

- Complete optical definitions: Line of Sight, Horizontal Reference Line, Observer Eye Height.
- Equivalence Theorem: Angle of Depression ≡ Ground Elevation via alternate interior angles.
- The 30°–60°–90° and 45°–45°–90° computational scaling engines.
- Canonical 2-triangle models, lake reflection optics, and uniform velocity motion.

How to use it for maximum marks

Always sketch a neat, labeled right-angled triangle diagram first (CBSE reserves 1 to 1.5 marks for the diagram alone). Explicitly declare the right triangle in text ("In right $\triangle ABC$, $\angle B = 90°$"). State the ratio symbolically before substituting values. Retain surds $(\sqrt{3}, \sqrt{2})$ in factored form throughout algebraic steps; substitute decimals only on the final line if requested.

Official syllabus scope (CBSE Class X 2026–27)

The published CBSE Class X curriculum for 2026–27 prescribes **Heights and Distances: Angle of elevation, Angle of Depression, simple problems on heights and distances**.
Strict Constraints: Problems should not involve more than **two right triangles**. Angles of elevation/depression are restricted exclusively to $\mathbf{30°}, \mathbf{45°}, \mathbf{60°}$.
Excluded Topics: 3D non-coplanar multi-plane systems, arbitrary non-standard angle tables $(15°, 75°)$, networks with ≥ 3 right triangles, and deleted complementary angle identities $(\tan(90° - \theta) = \cot\theta)$.

Source: CBSE Mathematics Class X Curriculum (2026–27), Unit V: Trigonometry.

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Prepared for learning and revision – not an official CBSE publication.

CHAPTER 9

CORE SYLLABUS

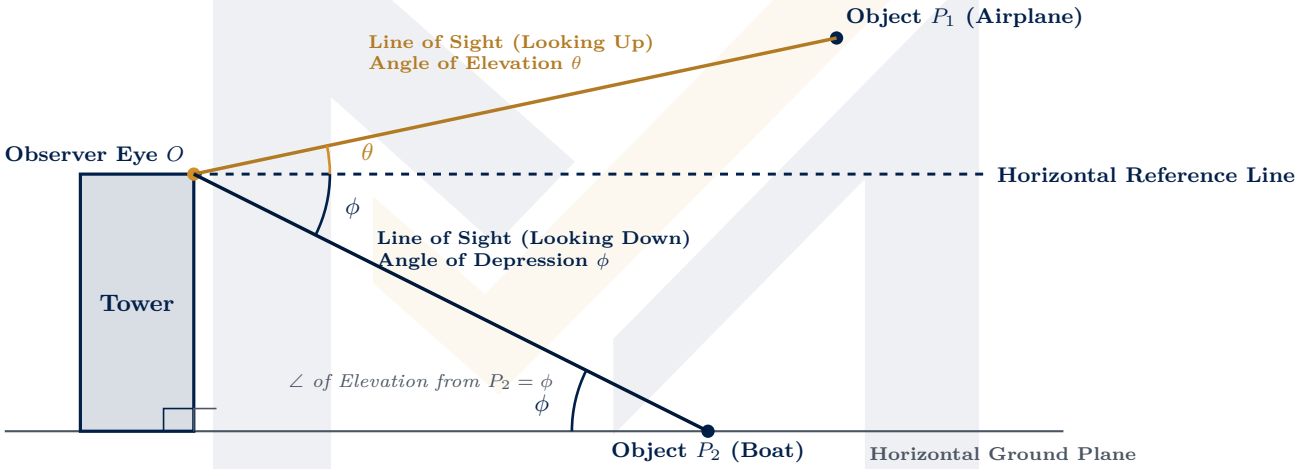
GEOMETRIC FOUNDATIONS

The Optics of Observation: Sight Lines, Elevation & Depression

Module 1 | Core Definitions & Physical Principles

Trigonometric heights and distances translate surveyor and eye observations into solvable Euclidean right-angled triangles.

- **Line of Sight:** The straight direct line drawn from the eye of the observer to the object viewed.
- **Horizontal Reference Line:** The straight horizontal line passing through the observer's eye, parallel to the ground surface.
- **Angle of Elevation (θ):** The angle formed by the line of sight with the horizontal reference line when the target object is **above** the horizontal eye level (i.e., looking **upward**). Domain: $0° < \theta < 90°$.
- **Angle of Depression (ϕ):** The angle formed by the line of sight with the horizontal reference line when the target object is **below** the horizontal eye level (i.e., looking **downward**). Domain: $0° < \phi < 90°$.



Module 2 | The Alternate Interior Angle Theorem (Z-Property)

Mathematical Equivalence of Depression and Ground Elevation

Since the observer's horizontal reference line is mathematically parallel to the horizontal ground plane:

$$\text{Horizontal Line through Eye} \parallel \text{Horizontal Ground Plane}$$

The line of sight acts as a transversal line cutting two parallel lines. By Euclid's Alternate Interior Angles Theorem:

$$\text{Angle of Depression } (\phi) = \text{Ground Angle of Elevation } (\phi)$$

Operational Board Exam Rule: When an angle of depression is stated, ALWAYS draw the dashed horizontal eye line first, mark ϕ, and then immediately transfer ϕ to the ground right-angled triangle.

Maths Manthan 3-Step Diagramming Protocol for Board Exams

1. **Perpendicular Anchor:** Draw the ground line and vertical objects with a right-angle ($90°$) symbol.
2. **Reference Sight Line:** For depression problems, sketch the horizontal line of sight before drawing the sight ray.
3. **Explicit Naming:** Label every vertex ($A, B, C \ldots$) and declare right triangles explicitly in your working.

CHAPTER 9

CORE SYLLABUS

OBSERVER MODELING

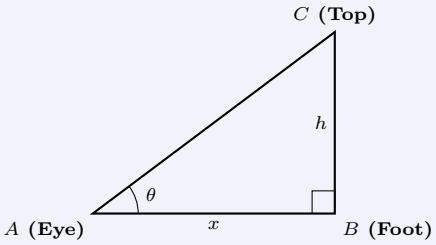
Point Observer vs. Observer with Height & Eye-Level Mechanics

Module 1 | The Geometry of Observer Height

CBSE problems categorize observers into two distinct geometrical models:

Model A: Point Observer (Height Ignored)

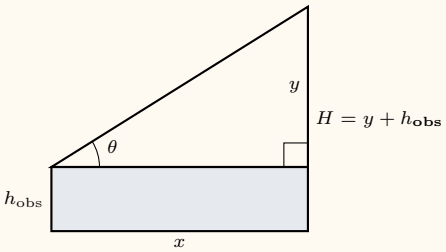
When the observer's physical height is not stated in the problem statement, the observer's eye is treated as a geometric point directly on the ground surface ($h_{\text{obs}} = 0$).



Total Object Height: $\mathbf{h = x \tan \theta}$.

Model B: Physical Observer with Height h_{obs}

When observer height is given (e.g., 1.5 m tall boy), the right triangle sits on a rectangle of height h_{obs}. The vertical leg of the triangle is the effective height y.



Total Height: $\mathbf{H = y + h_{obs} = x \tan \theta + h_{obs}}$.

Module 2 | Effective Vertical Leg Formulation

Effective Height Shift Formula

Let H be the total height of a vertical tower, and h_{obs} be the height of the observer.

$$\mathbf{h_{eff} = H - h_{obs} = x \tan \theta \iff H = x \tan \theta + h_{obs}}$$

Critical Warning: The right-angled triangle only models the space above the observer's eye line. Always add h_{obs} back to determine the final height of the building or chimney!

Module 3 | Worked Board Archetype: The 1.5-Meter Observer

Standard Board Problem: Observer Viewing a Chimney

Problem: An observer 1.5 m tall is 28.5 m away from a chimney. The angle of elevation of the top of the chimney from her eyes is 45°. What is the height of the chimney?

Solution:

Step 1: Set up the geometric model. Let AB be the chimney of total height H, and $CD = 1.5$ m be the observer. Draw horizontal line $DE \parallel CB$ meeting AB at E.

Then, $EB = CD = 1.5$ m, and $DE = CB = 28.5$ m. In right $\triangle ADE$, $\angle AED = 90°$ and $\angle ADE = 45°$.

Step 2: Calculate effective triangle height AE.

$$\tan 45° = \frac{AE}{DE} \implies 1 = \frac{AE}{28.5} \implies AE = 28.5 \text{ m}$$

Step 3: Calculate total chimney height H.

$$H = AB = AE + EB = 28.5 \text{ m} + 1.5 \text{ m} = \mathbf{30} \text{ m}$$

Final Statement: The total height of the chimney is **30** m.

CHAPTER 9

CORE SYLLABUS

THE COMPUTATIONAL ENGINE

Trigonometric Ratio Selection, Exact Surds & Geometric Scaling

Module 1 | The Ratio Selection Protocol: Deciding Between tan, sin, and cos

Trig Ratio	Formula	Typical Geometric Context	Board Frequency
Tangent ($\tan\theta$)	$\frac{\text{Opposite}}{\text{Adjacent}} = \frac{h}{x}$	Towers, buildings, statues, shadows, horizontal ground separations.	**≈ 85% of Exam Problems**
Sine ($\sin\theta$)	$\frac{\text{Opposite}}{\text{Hypotenuse}} = \frac{h}{L}$	Flying kites (taut string length), broken tree trunk bent to ground, ramps.	**≈ 12% of Exam Problems**
Cosine ($\cos\theta$)	$\frac{\text{Adjacent}}{\text{Hypotenuse}} = \frac{x}{L}$	Foot distance of leaning ladder from wall given physical ladder length.	**≈ 3% of Exam Problems**

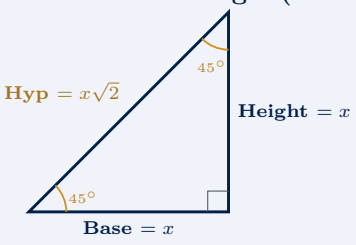
Module 2 | Exact Standard Trigonometric Ratios Table for Heights & Distances

Under the CBSE 2026–27 curriculum, only three special acute angles are permitted:

Angle θ	$\sin\theta$	$\cos\theta$	$\tan\theta$	$\cot\theta$	Geometric Interpretation
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\mathbf{1}}{\sqrt{\mathbf{3}}} = \frac{\sqrt{\mathbf{3}}}{\mathbf{3}}$	$\sqrt{\mathbf{3}}$	Base is $\sqrt{3} \approx 1.732$ times Height (Base > Height)
45°	$\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$	$\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$	**1**	**1**	Isosceles Right Triangle: Height = Base
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{\mathbf{3}}$	$\frac{\mathbf{1}}{\sqrt{\mathbf{3}}} = \frac{\sqrt{\mathbf{3}}}{\mathbf{3}}$	Height is $\sqrt{3} \approx 1.732$ times Base (Height > Base)

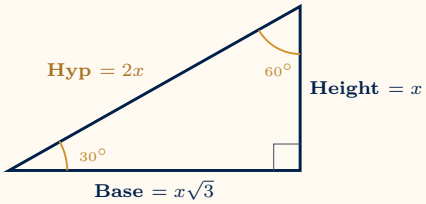
Module 3 | The Special Right Triangles & Rapid Scaling Laws

The 45°–45°–90° Triangle (Isosceles)



Side Ratio: $1 : 1 : \sqrt{2}$. Instant Rule: Height = Base.

The 30°–60°–90° Triangle (Scalene)



Side Ratio: $1 : \sqrt{3} : 2$. Opposite 30° is half hypotenuse.

Module 4 | Essential Algebraic Surd Tools & Rationalization

Golden Surd Identities for Error-Free Simplification

Never leave radicals in denominators: $\frac{\mathbf{1}}{\sqrt{\mathbf{3}}} = \frac{\sqrt{\mathbf{3}}}{\mathbf{3}}$, $\frac{\mathbf{1}}{\sqrt{\mathbf{3}}-\mathbf{1}} = \frac{\sqrt{\mathbf{3}}+\mathbf{1}}{\mathbf{2}}$, $\frac{\mathbf{1}}{\sqrt{\mathbf{3}}+\mathbf{1}} = \frac{\sqrt{\mathbf{3}}-\mathbf{1}}{\mathbf{2}}$.

Constants: $\sqrt{3} \approx 1.732$, $\sqrt{2} \approx 1.414$. Keep expressions factored throughout algebra; substitute decimals only on final step.

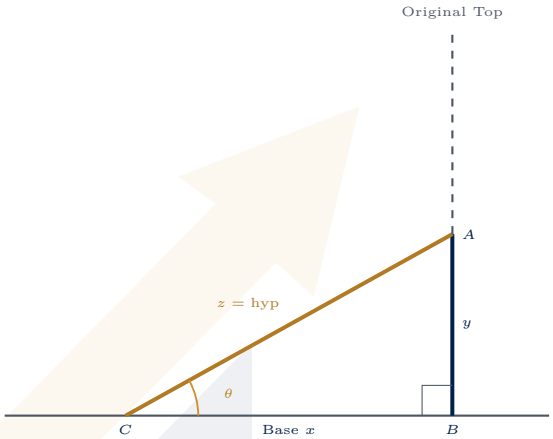
CHAPTER 9

WORKED MODEL

SINGLE-TRIANGLE ARCHETYPES

Broken Tree Stem, Inclined Kite String & Observer Models

Archetype 1 | The Broken Tree Model (Hypotenuse Fold-Over)



Pedagogical Insight: A tree of original vertical height H breaks at point A at height y above the ground. The upper broken stem of length z bends without detaching, touching the ground at point C at distance x from base B, forming angle θ.

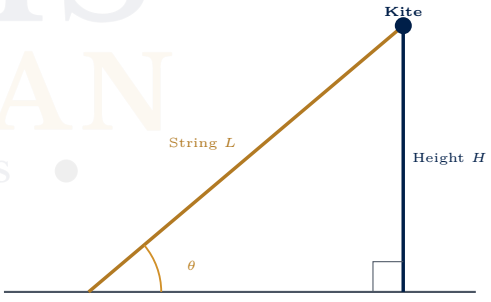
- Unbroken trunk: $AB = y$.
- Broken folded stem: $AC = z$.
- Ground distance: $BC = x$.
- Total original height: $\mathbf{H = y + z}$.

Governing Formulas & Worked Board Archetype (8m Tree)

$$\tan\theta = \frac{y}{x} \implies y = x\tan\theta, \qquad \cos\theta = \frac{x}{z} \implies z = \frac{x}{\cos\theta} \implies \mathbf{H = y + z = x\left(\frac{\sin\theta + 1}{\cos\theta}\right)}$$

Problem $(x = 8\,\mathbf{m}, \theta = 30°)$**:** $y = 8\tan 30° = \frac{8}{\sqrt{3}}$ m, $\quad z = \frac{8}{\cos 30°} = \frac{16}{\sqrt{3}}$ m $\implies H = \frac{24}{\sqrt{3}} = \mathbf{8\sqrt{3}}\,\text{m} \approx \mathbf{13.86}\,\text{m}$.

Archetype 2 | The Flying Kite with Incline & Taut String



Pedagogical Insight: A kite flies at a vertical altitude H attached to a string of length L inclined at angle θ to the horizontal. If held by an observer of height h_{obs}, the right triangle vertical opposite is $y = H - h_{\text{obs}}$.

$$\sin\theta = \frac{\mathbf{H - h_{obs}}}{\mathbf{L}} \implies \mathbf{L} = \frac{\mathbf{H - h_{obs}}}{\sin\theta}$$

If string is anchored directly on ground ($h_{\text{obs}} = 0$): $\mathbf{L} = \frac{\mathbf{H}}{\sin\theta}$.

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Worked Board Archetype: Kite at 60m Height

Problem: A kite is flying at a height of 60 m above the ground. The string attached is temporarily tied to a point on the ground at an inclination of 60°. Find the length of the string, assuming no slack.

Solution: In right $\triangle ABC$, $\sin 60^\circ = \frac{60}{L} \implies \frac{\sqrt{3}}{2} = \frac{60}{L} \implies L = \frac{120}{\sqrt{3}} = \frac{120\sqrt{3}}{3} = \mathbf{40\sqrt{3}}\,\text{m} \approx \mathbf{69.28}\,\text{m}$.

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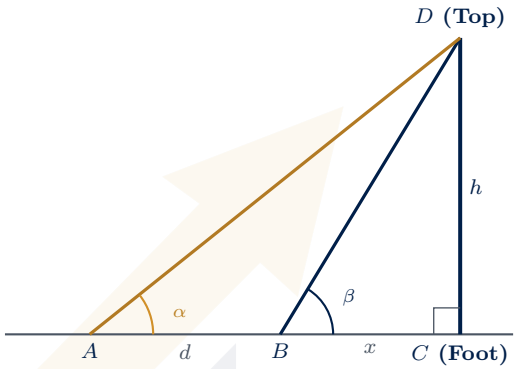
CHAPTER 9

CORE SYLLABUS

DUAL-TRIANGLE GEOMETRIES

Nested Same-Side Elevations & Shadow Lengthening Models

Configuration A | Dual Elevations on the Same Side (Moving Observer / Shadow)



Pedagogical Principle: An observer moves towards (or away from) a vertical tower $CD = h$ along a horizontal line. The angle of elevation changes from α to β as distance d is traversed. Note that as the observer moves closer, the angle strictly increases: $\beta > \alpha$.

- Tower height: $CD = h$.
- Near ground point: B, with elevation $\angle CBD = \beta$.
- Far ground point: A, with elevation $\angle CAD = \alpha$.
- Distance traversed: $AB = d$, near distance: $BC = x$.

Master Derivation for Moving Observer / Shadow Lengthening

In right $\triangle BCD$: $\cot\beta = \dfrac{x}{h} \implies x = h\cot\beta$. In right $\triangle ACD$: $\cot\alpha = \dfrac{x+d}{h} \implies x + d = h\cot\alpha$.
Subtracting the first equation from the second eliminates intermediate variable x:

$$\mathbf{d} = \mathbf{h}(\cot\alpha - \cot\beta) \iff \mathbf{h} = \frac{\mathbf{d}}{\cot\alpha - \cot\beta} = \frac{\mathbf{d}\tan\alpha\tan\beta}{\tan\beta - \tan\alpha}$$

Special Board Case ($\alpha = 30°, \beta = 60°$): $d = h\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right) = h\left(\frac{2}{\sqrt{3}}\right) \implies \mathbf{h = \frac{\sqrt{3}}{2}d}, \quad \mathbf{x = \frac{d}{2}}$.

Board Strategy Tip: Using $\cot\theta = \frac{\text{base}}{\text{height}}$ directly is faster because the unknown height h remains as a common single denominator.

Worked Board Archetype: Shadow Lengthening by 40 Meters

Problem: The shadow of a tower standing on level ground is found to be 40 m longer when the Sun's altitude is 30° than when it is 60°. Find the height of the tower.
Solution: Let h be the height of tower CD, and $x = BC$ be the shadow at 60°.

$$\text{In right } \triangle BCD: \ \tan 60° = \frac{h}{x} \implies \sqrt{3} = \frac{h}{x} \implies x = \frac{h}{\sqrt{3}}$$

$$\text{In right } \triangle ACD: \ \tan 30° = \frac{h}{x+40} \implies \frac{1}{\sqrt{3}} = \frac{h}{\frac{h}{\sqrt{3}} + 40} \implies \frac{h}{\sqrt{3}} + 40 = h\sqrt{3}$$

$$40 = h\sqrt{3} - \frac{h}{\sqrt{3}} = h\left(\frac{3-1}{\sqrt{3}}\right) = \frac{2h}{\sqrt{3}} \implies h = \frac{40\sqrt{3}}{2} = \mathbf{20\sqrt{3}}\,\text{m} \approx \mathbf{34.64}\,\text{m}$$

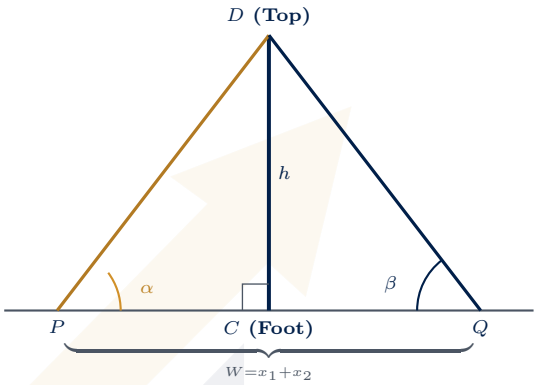
CHAPTER 9

CORE SYLLABUS

DUAL-TRIANGLE GEOMETRIES

Opposite-Side Observations & River Width Models

Configuration B | Observers on Opposite Sides of a Monument (River Crossing)



Pedagogical Principle: Two observation points P and Q lie on opposite sides of a central vertical tower or lighthouse $CD = h$ (e.g., opposite banks of a river or two ships on opposite sides of a light tower). The angles of elevation to the top are α and β.

- Monument height: $CD = h$.
- Distance of first point: $x_1 = PC$.
- Distance of second point: $x_2 = CQ$.
- Total separation distance: $\mathbf{W = PQ = x_1 + x_2}$.

Master Governing Formula for Opposite-Side Configurations

$$\text{In right } \triangle PCD: \ \cot\alpha = \frac{x_1}{h} \implies x_1 = h\cot\alpha, \qquad \text{In right } \triangle QCD: \ \cot\beta = \frac{x_2}{h} \implies x_2 = h\cot\beta$$

$$\mathbf{W = x_1 + x_2 = h}(\cot\alpha + \cot\beta) = \mathbf{h}\left(\frac{\tan\alpha + \tan\beta}{\tan\alpha\tan\beta}\right)$$

Comparison Rule: Same-side points **subtract** distances: $d = h(\cot\alpha - \cot\beta)$. Opposite-side points **add** distances: $W = h(\cot\alpha + \cot\beta)$.

Worked Board Archetype: Two Ships on Opposite Sides of a Lighthouse

Problem: As observed from the top of a 75 m high lighthouse from the sea-level, the angles of depression of two ships on opposite sides of the lighthouse are 30° and 45°. Find the distance between the two ships.

Solution:

Step 1: Declare the geometry. Let lighthouse $CD = 75\,\text{m}$. Let P and Q be the two ships on opposite sides. Angles of depression equal angles of elevation $\angle DPC = 30°$ and $\angle DQC = 45°$ by alternate interior angles.

Step 2: Distance to ship P. In right $\triangle PCD$:

$$\tan 30° = \frac{CD}{PC} \implies \frac{1}{\sqrt{3}} = \frac{75}{x_1} \implies x_1 = 75\sqrt{3}\,\text{m}$$

Step 3: Distance to ship Q. In right $\triangle QCD$:

$$\tan 45° = \frac{CD}{QC} \implies 1 = \frac{75}{x_2} \implies x_2 = 75\,\text{m}$$

Step 4: Total distance W between ships.

$$W = x_1 + x_2 = 75\sqrt{3} + 75 = \mathbf{75}(\sqrt{\mathbf{3}} + \mathbf{1})\,\text{m} \approx 75(2.732) = \mathbf{204.9}\,\text{m}$$

CHAPTER 9

CORE SYLLABUS

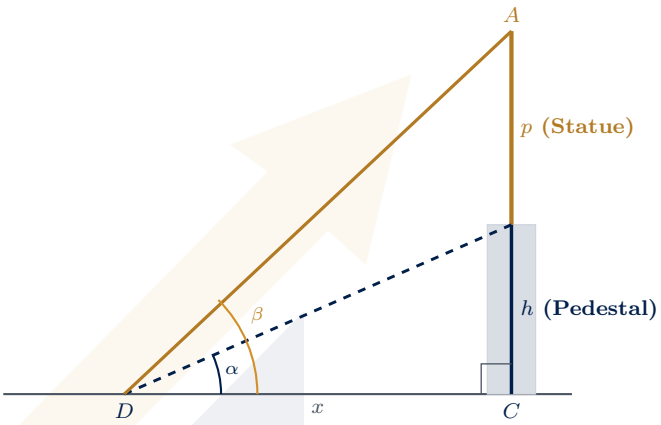
MULTI-LEVEL SYSTEMS

Pedestals, Flagstaffs & Vertical Stacking Geometries

Configuration C | Statue or Flagstaff Atop a Pedestal / Building

Pedagogical Principle: A statue or flagstaff of height p sits atop a pedestal/building of height h. From a single ground observation point D at distance x, the angles of elevation to the bottom and top of the statue are α and β ($\beta > \alpha$).

- Lower pedestal: $BC = h$.
- Upper statue: $AB = p$. Total height: $AC = h + p$.
- Common horizontal base: $CD = x$.



Master Governing Formulas for Pedestal & Statue

$$\text{In } \triangle BCD: \ x = \frac{h}{\tan\alpha} = h\cot\alpha. \qquad \text{In } \triangle ACD: \ x = \frac{h+p}{\tan\beta}$$

Equating the common base x yields the direct closed-form formulas:

$$\mathbf{p = x(\tan\beta - \tan\alpha) = h\left(\frac{\tan\beta}{\tan\alpha} - 1\right) \iff h = \frac{p\tan\alpha}{\tan\beta - \tan\alpha}}$$

Worked Board Archetype: Statue of Height 1.6m on Pedestal

Problem: A 1.6 m tall statue stands on the top of a pedestal. From a point on the ground, the angle of elevation of the top of the statue is 60° and from the same point the angle of elevation of the top of the pedestal is 45°. Find the height of the pedestal.

Solution: Let pedestal height be $h = BC$, statue $p = AB = 1.6\,\text{m}$, and base $CD = x$.

$$\text{In right } \triangle BCD: \ \tan 45^\circ = \frac{h}{x} \implies 1 = \frac{h}{x} \implies x = h$$

$$\text{In right } \triangle ACD: \ \tan 60^\circ = \frac{h+1.6}{x} \implies \sqrt{3} = \frac{h+1.6}{h} \implies h\sqrt{3} = h + 1.6$$

$$h(\sqrt{3}-1) = 1.6 \implies h = \frac{1.6}{\sqrt{3}-1} = \frac{1.6(\sqrt{3}+1)}{3-1} = \mathbf{0.8(\sqrt{3}+1)}\,\text{m} \approx 0.8(2.732) = \mathbf{2.186}\,\text{m}$$

Common Board Exam Pitfall on Pedestals

Students frequently make the fatal mistake of assuming $\triangle ABD$ is a right-angled triangle. It is NOT! The right angle is strictly at the ground base C. Always formulate equations using right $\triangle BCD$ and right $\triangle ACD$.

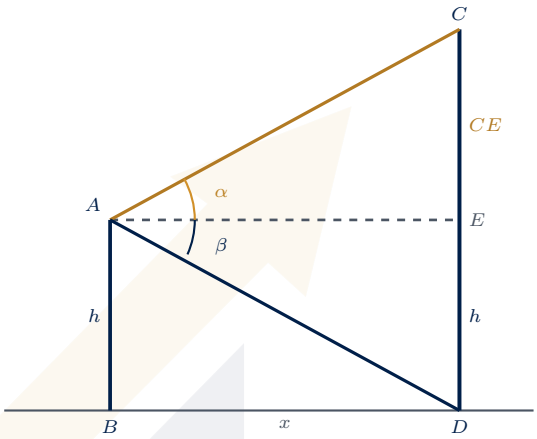
CHAPTER 9

CORE SYLLABUS

MULTI-LEVEL SYSTEMS

Building-to-Tower: Elevation & Depression Combinations

Configuration D | Building to Adjacent Tower (Elevation + Depression Dual)



Pedagogical Principle: An observer standing on top of a building of height h views an adjacent taller cable tower:

- Looking **up** at the top: Angle of elevation $= \alpha$.
- Looking **down** at the foot: Angle of depression $= \beta$.
- Rectangle formed: $ABDE$ has height h and width x.
- Tower height: $\mathbf{H = CD = CE + h}$.

Governing Formula for Building-to-Tower Observation

$$\text{In right } \triangle ADE: \ \tan\beta = \frac{DE}{AE} = \frac{h}{x} \implies x = \frac{h}{\tan\beta} = h\cot\beta$$

$$\text{In right } \triangle ACE: \ CE = x\tan\alpha = (h\cot\beta)\tan\alpha = h\frac{\tan\alpha}{\tan\beta}$$

$$\mathbf{H_{tower} = CD = CE + ED = h\left(1 + \frac{\tan\alpha}{\tan\beta}\right) = h(1 + \tan\alpha\cot\beta)}$$

Worked Board Archetype: 7m Building and Cable Tower

Problem: From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 45°. Determine the height of the tower.

Solution:

Step 1: Set up rectangle and right triangles. Let $AB = 7$ m be the building and CD be the cable tower. Draw horizontal line $AE \perp CD$. Then $AB = ED = 7$ m and $AE = BD = x$.

Step 2: Find width x from angle of depression. In right $\triangle ADE$:

$$\tan 45^\circ = \frac{DE}{AE} \implies 1 = \frac{7}{x} \implies x = 7\,\text{m}$$

Step 3: Find upper segment CE from angle of elevation. In right $\triangle ACE$:

$$\tan 60^\circ = \frac{CE}{AE} \implies \sqrt{3} = \frac{CE}{7} \implies CE = 7\sqrt{3}\,\text{m}$$

Step 4: Total height of the cable tower.

$$H = CD = CE + ED = 7\sqrt{3} + 7 = \mathbf{7(\sqrt{3} + 1)}\,\text{m} \approx 7(2.732) = \mathbf{19.124}\,\text{m}$$

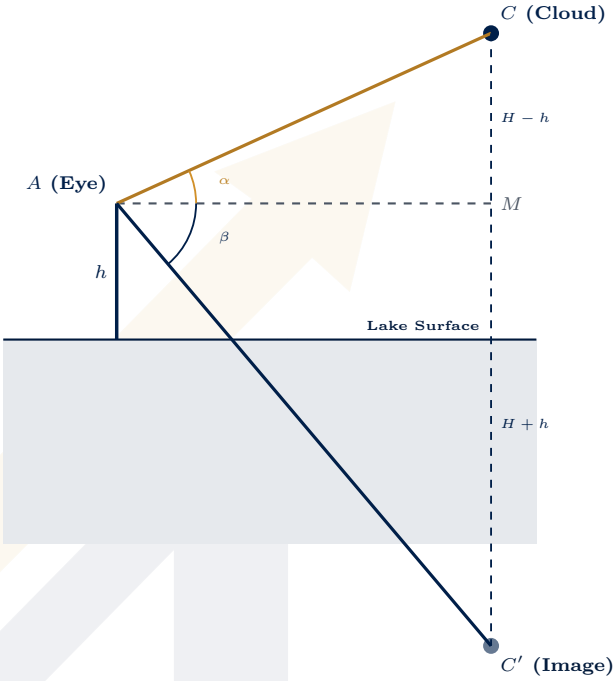
CHAPTER 9

WORKED MODEL

ADVANCED BOARD MODELS (HOTS)

Lake Reflection Optics: Real Cloud vs. Virtual Image

HOTS Archetype 1 | The Cloud and its Reflection in a Lake



Optical Law in Trigonometry: By specular reflection from a calm lake surface, the depth of the virtual image (C') beneath the water level is strictly equal to the height of the real cloud (C) above the water level:

$$\text{Cloud Height } \mathbf{H} = \text{Virtual Image Depth } \mathbf{H}$$

Let the observer be at platform height h above water level.

- Real cloud altitude above observer: $CM = H - h$.
- Virtual image depth below observer: $C'M = H + h$.
- Horizontal distance to cloud: $AM = x$.

Closed-Form Derivation for Cloud Altitude above Water

$$\text{In right } \triangle ACM: \ x = \frac{H-h}{\tan\alpha}, \qquad \text{In right } \triangle AC'M: \ x = \frac{H+h}{\tan\beta}$$

Equating expressions for horizontal separation x:

$$\frac{H-h}{\tan\alpha} = \frac{H+h}{\tan\beta} \implies (H-h)\tan\beta = (H+h)\tan\alpha \iff \mathbf{H} = \mathbf{h}\left(\frac{\tan\beta+\tan\alpha}{\tan\beta-\tan\alpha}\right)$$

Worked Board Archetype: Cloud at 60m Platform Height

Problem: The angle of elevation of a cloud from a point 60 m above a lake is 30° and the angle of depression of its reflection in the lake is 60°. Find the height of the cloud above the lake.
Solution: Let H be cloud height above lake, and $h = 60$ m. In $\triangle ACM$ and $\triangle AC'M$:

$$\tan 30^\circ = \frac{H-60}{x} \implies x = (H-60)\sqrt{3}, \qquad \tan 60^\circ = \frac{H+60}{x} \implies x = \frac{H+60}{\sqrt{3}}$$

$$(H-60)\sqrt{3} = \frac{H+60}{\sqrt{3}} \implies 3(H-60) = H+60 \implies 2H = 240 \implies \mathbf{H = 120\,m}$$

Final Statement: The height of the cloud above the lake surface is **120** m.

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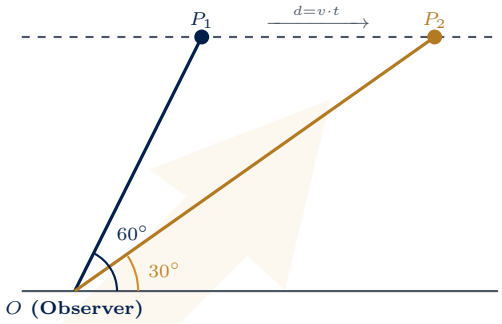
CHAPTER 9

WORKED MODEL

ADVANCED BOARD MODELS (HOTS)

Uniform Velocity Mechanics: Moving Aircraft & Boats

HOTS Archetype 2 | Uniform Velocity & Rate of Motion (Aircraft / Approaching Boat)



Physical Principle: An object moves along a straight horizontal path at constant speed v. The observation angle changes as time elapses:

- Uniform velocity equation: $\mathbf{\Delta x = v \cdot t}$.
- Constant altitude of airplane: H.
- Initial elevation angle at t_1: $\theta_1 = 60°$.
- Final elevation angle after time t: $\theta_2 = 30°$.
- Distance traversed: $\mathbf{d = x_2 - x_1}$.
- Uniform speed: $\mathbf{v = \frac{d}{t}}$.

Governing Mathematical Derivation for Cruising Airplane

$$\text{Initial base distance: } x_1 = H\cot 60° = \frac{H}{\sqrt{3}}, \qquad \text{Final base distance: } x_2 = H\cot 30° = H\sqrt{3}$$

$$\text{Horizontal distance covered in time } t: \; d = x_2 - x_1 = H\sqrt{3} - \frac{H}{\sqrt{3}} = \frac{2H}{\sqrt{3}}$$

$$\text{Speed } \mathbf{v} = \frac{\mathbf{d}}{\mathbf{t}} = \frac{\mathbf{2H}}{\sqrt{\mathbf{3}}\,\mathbf{t}} \;\text{[m/s]} \qquad \Longleftrightarrow \qquad \mathbf{v}_{\text{km/h}} = \mathbf{v} \times \frac{\mathbf{18}}{\mathbf{5}} \;\text{[km/h]}$$

Worked Board Archetype: Airplane at Altitude $1500\sqrt{3}$ Meters

Problem: An aeroplane at an altitude of $1500\sqrt{3}$ m observes that the angle of elevation changes from 60° to 30° in 15 seconds while flying horizontally. Find the speed of the aeroplane in km/h.

Solution:

Step 1: Compute initial and final base distances.

$$x_1 = \frac{1500\sqrt{3}}{\tan 60°} = \frac{1500\sqrt{3}}{\sqrt{3}} = 1500\,\text{m}, \qquad x_2 = \frac{1500\sqrt{3}}{\tan 30°} = \frac{1500\sqrt{3}}{1/\sqrt{3}} = 4500\,\text{m}$$

Step 2: Compute distance traversed in 15 seconds.

$$d = x_2 - x_1 = 4500\,\text{m} - 1500\,\text{m} = 3000\,\text{m}$$

Step 3: Calculate speed and convert to km/h.

$$v = \frac{d}{t} = \frac{3000\,\text{m}}{15\,\text{s}} = 200\,\text{m/s} \implies \mathbf{v = 200 \times \frac{18}{5} = 40 \times 18 = 720}\,\text{km/h}$$

CHAPTER 9

REFERENCE MATRIX

QUICK-REVISION MATRIX

Comprehensive Canonical Geometries & Rapid Shortcuts

Module 1 | Master Trigonometric Geometry Reference Matrix

Geometric Configuration	Primary Unknown	Governing Master Formula	10-Second Mental Check
Single Triangle	Height / Distance	$h = x\tan\theta \iff x = h\cot\theta$	At 45°, $h = x$; at 30°, $x > h$.
Broken Tree Stem	Total Tree Height	$H = y + z = x\left(\dfrac{1+\sin\theta}{\cos\theta}\right)$	For 30°, $H = x\sqrt{3}$.
Nested Same-Side	Distance Traversed	$d = h(\cot\alpha - \cot\beta)$	For $30^\circ, 60^\circ$, $h = \dfrac{\sqrt{3}}{2}d$ and $x = \dfrac{d}{2}$.
Opposite-Side River	Total Separation	$W = h(\cot\alpha + \cot\beta)$	Always strictly greater than same-side.
Statue on Pedestal	Pedestal Height	$h = \dfrac{p\tan\alpha}{\tan\beta - \tan\alpha}$	If $\alpha = 45^\circ, \beta = 60^\circ$, $h = \dfrac{p}{\sqrt{3}-1}$.
Building to Tower	Total Tower Height	$H = h(1 + \tan\alpha\cot\beta)$	If $\alpha = 60^\circ, \beta = 45^\circ$, $H = h(\sqrt{3}+1)$.
Cloud in Lake	Real Cloud Height	$H = h\left(\dfrac{\tan\beta + \tan\alpha}{\tan\beta - \tan\alpha}\right)$	If $\alpha = 30^\circ, \beta = 60^\circ$, $H = 2h$.

Module 2 | Supporting Symmetric Configuration: Two Poles of Equal Height

Board Model: Two Poles on Opposite Sides of an 80m Road

Configuration: Two poles of equal height h stand on opposite sides of a road of width $W = 80\,\text{m}$. From a point between them, angles of elevation of the tops are 60° and 30°.

$$\tan 60^\circ = \frac{h}{x} \implies h = x\sqrt{3}, \qquad \tan 30^\circ = \frac{h}{80-x} \implies h = \frac{80-x}{\sqrt{3}}$$

$$x\sqrt{3} = \frac{80-x}{\sqrt{3}} \implies 3x = 80 - x \implies 4x = 80 \implies \mathbf{x = 20}\,\text{m}$$

$$\mathbf{h = 20\sqrt{3}}\,\text{m}, \qquad \text{Distances from the point: } \mathbf{20}\,\text{m and } \mathbf{60}\,\text{m}.$$

Module 3 | Classical Geometric Extension: Complementary Angles ($h = \sqrt{ab}$)

Complementary Observation Points Shortcut

If angles of elevation of top of a tower from two ground points at distances a and b from its base are **complementary** ($\alpha + \beta = 90^\circ$):

$$\tan\alpha = \frac{h}{a}, \qquad \tan\beta = \tan(90^\circ - \alpha) = \cot\alpha = \frac{h}{b} \implies \tan\alpha \cdot \cot\alpha = \frac{h^2}{ab} = 1 \implies \mathbf{h = \sqrt{ab}}$$

CHAPTER 9

EXAM STRATEGY

BOARD STRATEGY & AUDIT

Canonical Geometries, Examiner Traps & Pre-Submission Checklist

Module 1 | The Five Lethal CBSE Board Exam Traps

Examiner Trap Alerts (Where Students Lose Marks)

1. **The Vertical Reference Line Blunder:** In depression problems, students often take the angle with the vertical wall. Remember: Angles of elevation and depression are **ALWAYS** measured against the horizontal reference line!
2. **The Vanishing Observer Height:** For a 1.5 m observer looking at a 30 m chimney, triangle leg is $30 - 1.5 = 28.5$ m. Forgetting to subtract initially (or forgetting to add 1.5 m back at the end) loses 2 marks.
3. **Premature Decimal Substitution:** Replacing $\sqrt{3}$ with 1.732 early creates rounding errors. Factor common terms, simplify fractions completely, and substitute decimals only on the last line.
4. **Missing Alternate Interior Angle Line:** When an angle of depression is given, failing to draw the horizontal line from the observer's eye is penalized under CBSE step-marking criteria.
5. **Speed Unit Mismatch:** Calculating airplane speed as distance (meters)/time (seconds) yields m/s. Failing to convert to km/h (multiply by 18/5) when asked costs the final accuracy mark.

Module 2 | Rapid 15-Second Arithmetic Verification Hacks

Instant Mental Sanity Checks

- **Distance-Angle Monotonicity:** As an observer approaches a tower, the elevation angle must strictly increase ($30^\circ \to 45^\circ \to 60^\circ$). If your calculation yields a smaller angle closer to the tower, reverse your geometry!
- **Isosceles Right Triangle Check:** At 45°, height must exactly equal base.
- **Hypotenuse Superiority:** The hypotenuse is strictly the longest side ($L > h$ and $L > x$). If string length L is less than height H, an algebraic error has occurred.

Module 3 | Five-Point Pre-Submission Board Exam Audit Checklist

Maths Manthan 5-Point Pre-Submission Audit Checklist

[ ] **1. Right-Angle Symbol & Labeled Vertices:** Is every vertical object marked perpendicular (90°) to the ground, and are all vertices named ($A, B, C \ldots$)?

[ ] **2. Triangle Declaration in Text:** Is each calculation preceded by a clear declaration (e.g., "In right-angled triangle ABC, $\angle B = 90^\circ$")?

[ ] **3. Depression Angle Construction:** Did I draw the dashed horizontal eye-line and mark alternate interior angles?

[ ] **4. Conjugate Rationalization:** Are all denominators cleared of radicals (e.g., multiplied by $\sqrt{3}+1$)?

[ ] **5. Physical Units:** Does every final answer include proper units ($\text{m}, \text{cm}, \text{km/h}$) highlighted in a neat box?

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End of Chapter 9 Handbook – Designed for Academic Excellence in CBSE Board Examinations.